What's new

# An experimental AI-math challenge: using tenth discriminant bounds to explore arguments ruling out Siegel zeroes

Terence Tao · 20 October 2026

https://terrytao.wordpress.com/2026/10/20/an-experimental-ai-math-challenge-using-tenth-discriminant-bounds-to-explore-arguments-ruling-out-siegel-zeroes/

tao2typ · converted to Typst 2026-10-21 (UTC)

---

In the spirit of my recent "Math 2.0" talk on new ways to use AI assistance to enhance the human understanding of mathematics, rather than substitute for it, I am using this post to propose an experimental challenge, inspired by the recent activity around excluding Landau–Siegel zeroes, as discussed in this previous post.

There is a well-known connection between Landau–Siegel zeroes and the tenth discriminant problem, which I will briefly recall here; an in-depth discussion on this problem can be found in this survey of Shafarevich. As far back as the work of Gauss, it was observed that there were nine discriminants $-d$ whose rings of integers for the associated quadratic field $\mathbf{Q}(\sqrt{-d})$ enjoyed unique factorization, namely

$$d = 1, 2, 3, 7, 11, 19, 43, 67, 163.$$

Gauss believed this list of nine to be the complete list of such discriminants (the "class number one problem"), but this was only proven by Heegner in 1952 by introducing the concept of a Heegner point on the modular curve; and the above numbers are now known as Heegner numbers.

Prior to Heegner's solution was the work of Heilbronn and Linfoot in 1934, who used what we now call the Deuring–Heilbronn repulsion phenomenon to show that the existence of a tenth discriminant in the above list can be used to "repel" any further such discriminants, so that there was at most one further Heegner number beyond the nine already known. However, this argument was ineffective: no upper bound on this putative tenth discriminant was given, and so there was no obvious way to solve the class number one problem purely from this result.

There is a close connection to Landau–Siegel zeroes: if $d$ is a Heegner number, and $\chi$ is the quadratic character associated to this quadratic field, then the class number formula gives $L(1,\chi) = \frac{\pi}{\sqrt{d}}$, and with a bit more work one can produce a Landau–Siegel zero $L(\beta,\chi) = 0$ with $1 - \beta \asymp \frac{1}{\sqrt{d}}$. For sufficiently large $d$, this is inconsistent with Siegel's theorem; but the standard proofs of this theorem also relied on the repulsion phenomenon and were also ineffective. It was only in 1976 that Goldfeld gave an effective version of Siegel's theorem that was enough to at least bound the tenth discriminant, given a suitable high rank elliptic curve. In followup work of Oesterlé, combined with a high rank elliptic curve of Gross and Zagier, the bound $d \leq e^{7000}$ was obtained; using a later elliptic curve of Brumer and Kramer, proven to have high rank

by Mestre, Oesterlé, and Serre, the bound was improved to $d \leq e^{55}$. See this survey of Goldfeld for more discussion.

But now we have a fair number of new (and effective) ways to rule out Siegel zeroes, either by establishing a quasi-Riemann hypothesis, or by some version of the argument in this previous post. In principle, each version of these arguments should also give a bound on the tenth discriminant $d$ (without using Heegner's solution to the class number one problem that rules out this discriminant). The task of making each of these estimates explicit enough to create such a bound is rather tedious for humans. However, it seems eminently suitable for AI.

Thus, the challenge I pose is, for each of the new manuscripts or digestions relating to either the quasi-Riemann hypothesis or the Landau–Siegel zero problem (or by some novel argument), to use an AI tool to work out an effective upper bound for a hypothetical tenth discriminant $d$, without using Heegner's methods, and to make best efforts to optimize this constant. This serves several purposes:

- If the threshold is low enough to resolve the remaining discriminants by direct computation, this would yield a new solution of the class number one problem — possibly one that is formalizable in Lean in a high quality fashion (suitable for instance for upstreaming to Mathlib).
- By performing this exercise on each of the different writeups, we obtain a way to compare the "strength" of each such writeup (after performing whatever optimizations are available for each writeup), which could give insight as to the relative advantages or disadvantages of different arrangements of the methods.
- The exercise of minimizing the upper bound may spur the creation of entirely new arguments to solve either the class number one problem, or the elimination of Landau–Siegel zeroes.
- There are many previous "failed" attempts at excluding the Landau–Siegel zero. It is possible that some of them contain ideas that can be combined with the newer solutions to this problem to "repair" the proofs, or at least improve the bound on the tenth discriminant.
- Efforts here may also help improve the state of knowledge on higher class number problems, which by the work of Watkins in 2004 is currently solved up to class number 100.

Furthermore, as both of these problems are already solved (for over seven decades, in the case of the class number one problem), there is little risk of such a competition inadvertently disrupting an ongoing research project.

I am therefore inviting interested readers to contribute in the comments any effective bounds on the tenth discriminant they can extract from either an existing argument, or a novel one, with any amount of AI use permitted for this challenge so long as it is disclosed. Note that the challenge is broader than simply getting the "best" bound overall amongst all possible approaches; I am also interested in what bounds one can extract from any specific approach, even if they are worse than, say, the bounds of Goldfeld or Oesterlé. In particular, "negative" results in which the bounds obtained from a given method end up being rather large are still scientifically useful pieces of information that are worth reporting. One could also explore other directions, such as bounds that avoid using extensive compute to squeeze out the maximal amount of optimization, but are instead clean enough that their arguments are human-verifiable.

Any large outputs should be posted as links rather than as plain text; for instance, pointing to a Github repository containing one's work would be a reasonable way to submit an entry.

I would ask contributors to not prioritize speed above quality; this is not intended to be a race regarding who is "first" to claim any given bound. The spirit is intended to be collaborative rather than competitive: think of a Polymath style project, updated for the AI era. In particular, as with traditional mathematical research, it would be appropriate to build upon previous contributions to obtain an improved bound for a given proof technique. (But in such cases, one should of course give appropriate attribution to the prior source.)

I will not insist on formal verification (e.g., in Lean) of the results, though I would ask anyone submitting a purely AI-generated solution to do enough due diligence to check the output (e.g., with an independent AI tool) before posting it here.

I am looking forward to the results of this experimental challenge, both for the mathematical insights gained, and for the lessons learned on how to crowdsource AI assistance in mathematics in a responsible fashion.

## Comments (26)

 · 20 October 2026

It is good to see that you are back. Thanks a lot!

 **Lior Silberman** · 20 October 2026

Dumping here some calculations from late last night. If $\mathbb{Q}(\sqrt{-d})$ has class number 1 then every rational prime $p < d/4$ is inert in the extension so has $\chi_d(p) = -1$. Letting $q$ be the conductor of the quadratic character (at most four times the discriminant), we examine

$$\sum_{\substack{p<q/16 \\ \chi_d(p)=-1}} \frac{\log p}{p} = \sum_{p<q/16} \frac{\log p}{p} - \sum_{p \mid q} \frac{\log p}{p}.$$

Suppose Mertens's estimate reads $\left|\sum_{p<x} \frac{\log p}{p} - \log x\right| \le C_{\mathrm{M}}$ Then

$$\sum_{p<q/16} \frac{\log p}{p} - \sum_{p \mid q} \frac{\log p}{p} \ge \log \tfrac{q}{16} - c_{\mathrm{M}} - \log\log q - (1 + c_{\mathrm{M}})$$

So a no-Landau–Siegel zeroes upper bound of the form

$$\sum_{\substack{p<x \\ \chi_d(p)=-1}} \frac{\log p}{p} \le \alpha \log x + \beta \log q + \log\log x + C_{\mathrm{LS}}$$

translates to the inequality (forfeiting a small gain)

$$\log q - \log\log q - (1 + 2C_{\mathrm{M}} + \log 16) \le (\alpha + \beta)\log q - \alpha \log 16 + \log\log q + C_{\mathrm{LS}}$$

that is

$$(1 - \alpha - \beta)\log q \le 2\log\log q + C_{\mathrm{LS}} + 2C_{\mathrm{M}} + (1 - \alpha)\log 16$$

which gives an upper bound on $q$ as long as $\alpha + \beta < 1$. For example if $\alpha = \frac{1}{2}, \beta = \frac{1}{4}, C_{\mathrm{M}} = 2, C_{\mathrm{LS}} = 10$ one gets about $q < 10^{15}$.

 **Lior Silberman** · 20 October 2026

It seems I made an error last night; I'm now getting a bound of $q < 10^{45}$. I also lost a $+1$ in the RHS of the last equation.

**Aleksei Kulikov** · 20 October 2026

I am sorry, as this is not related to the main point of the post, but I believe your history of effective versions of the Siegel theorem is grossly incorrect. What I believe to be correct I took from Goldfeld's own survey of the problem from 1985 https://pubs.ams.org/BULL/1985-13-01/S0273-0979-1985-15352-2

In his 1976 paper Goldfeld proved a lower bound on the class number in terms of the analytic rank of an elliptic curve. However, for his method to be actually useful he needed an elliptic curve with analytic rank at least 3, which was not known at that time. The first provable example of such an elliptic curve, and thus the first effective estimate, was established only in 1983 in the celebrated work of Gross and Zagier (in fact, when I read about the class number problem in the past, the pendulum of misattributing history was usually swinging in the other way, with calling it a "Gross–Zagier theorem", and not mentioning Goldfeld at all).

As far as I can see, all explicit constants actually come from the work of Oesterlé, the constant 7000 comes from optimizing the work of Goldfeld with the elliptic curve of Gross and Zagier (there is actually an extra small term depending on the prime factorization of the potential d, which might make the constant even larger, but since for class number 1 problem I believe we can restrict to prime discriminants, this is not a big issue). The constant 55 comes from a different elliptic curve, found by Brumer and Kramer, but, if I understood Goldfeld correctly, it was proved to have analytic rank 3 only later in the works of Mestre, Oesterlé and Serre.



**Terence Tao** · 20 October 2026

Thanks for the additional references and context! I have updated that portion of the blog post accordingly.

**twinheterodox** · 20 October 2026

$q < 10^{10}$, relying on Banks's Theorem 1.1.

I used codex to prepare the argument, running GPT-6-Astra max. Codex also generated the following summary.

Proof, source code, and verification outputs on GitHub

The analytic step is a classical theta-integral argument with explicit constants. Under the class-number-one assumption, for $q \geq 10^{10}$ it produces a real zero $\beta \in (0, 1)$ satisfying

$$1 - \beta \leq \frac{10}{\sqrt{q}},$$

and consequently

$$(1 - \beta) \log q \leq \frac{10 \log q}{\sqrt{q}} \leq 0.0024.$$

This contradicts Banks's stated lower bound

$$(1 - \beta) \log q \geq 0.003.$$

Below this threshold, an exact integer sieve uses the necessary condition that every prime smaller than $q/4$ is inert. Using primes only up to 109, it leaves precisely the absolute fundamental discriminants

$3, 4, 7, 8, 11, 19, 43, 67, 163.$

Two implementations using different algorithms agree, and reduced-form enumeration checks the surviving fields. The argument avoids Heegner's methods and does not use Banks's separate small-conductor computational extension.

AI disclosure: Codex produced this summary, the derivation, the write-up, and both programs. Both programs were written by the same assistant, and their results have been cross-checked. The mathematical argument was also cross-checked with a different frontier model. I would particularly welcome scrutiny of the theta-integral lemma and the application of Banks's theorem.

(Ed: To be clear: "I" in that last sentence is the AI model.)



**twinheterodox** · 20 October 2026

Ed: I will do improvements within the same general approach as replies rather than top-level comments. Codex summary follows.

$q < 5 \times 10^7$, improving the previous bound by a factor of 200.

Version 2: complete proof, source code, and verification outputs. The original version remains available separately.

There are two changes. First, a sharper theta-integral estimate, retaining the terms from $\pm 1, \ldots, \pm 30$, gives

$$1 - \beta < \frac{2}{\sqrt{q}}$$

for $q \geq 49,000,000$, under the class-number-one assumption.

Second, using Banks's Lemmas 4.2 and 5.1 and Corollary 6.3 with $H = 10^9$ and $N = q^{18}$ gives

$$(1 - \beta) \log q > 0.00509$$

for $q \geq 5 \times 10^7$. The sharper theta estimate gives

$$(1 - \beta) \log q < 0.00502$$

throughout the same range, yielding a contradiction. These intermediate results of Banks remain external inputs. His separate small-conductor computation is not used.

The original finite sieve already covers the larger range, so its code, outputs, and final list of nine discriminants are unchanged. Version 2 contains the complete cumulative proof and a certificate with 23 exact rational checks of the new constants.

AI disclosure: Codex generated the refinement, the write-up, the arithmetic checker, and this summary.

**twinheterodox** · 20 October 2026

Ed: the manuscripts for each version are getting unwieldy (and unreadable for me) since they're relying implicitly on prior versions. I have codex rewriting this version 3 proof to be more self-contained. It should be finished in a few minutes. Codex summary follows:

$q < 10^6$, a further factor of 50 improvement over v2.

Version 3: proof, source code, and verification outputs. Earlier versions remain available separately.

This continues to use Banks's determinant construction, with a different input from the class-number-one assumption: a direct estimate for split primes replaces the theta-integral zero construction.

Under that assumption, each split prime supplies two primitive lattice points of positive imaginary part with that prime as norm. Counting these points, and sieving out pairs whose coordinates share a factor among $2, 3, 5, 7$, gives

$$\sum_{\substack{p \le X \\ \chi_D(p)=1}} \frac{\log p}{p} < \frac{0.493(\log X)^2 + 5.04 \log q}{\sqrt{q}}$$

for $q \ge 10^6$ and $X \ge q/4$. Combined with explicit prime estimates of Rosser-Schoenfeld and a bound on ramified primes, this supplies a lower bound for the weighted sum of inert primes.

Applying Banks's Lemmas 4.2, 5.1, and 6.2 with $H = 10^9$, $N = q^{13}$, and $M = N^{4/3}$ then gives a contradiction throughout $q \ge 10^6$. The final comparison has an exact rational margin greater than $0.022$.

The determinant construction and cited lemmas remain external inputs. This argument uses Banks's Lemma 6.2 directly, without the zero-dependent prime-bias estimate or Corollary 6.3. No Heegner method or small-conductor zero computation is used.

The original finite sieve already covers the remaining range, so its code, outputs, and final list of nine discriminants are unchanged. The new checker passes 26 exact arithmetic checks and 22 finite checks of the lattice formulas and prime-norm multiplicities.

AI disclosure: Codex generated the new argument, the write-up, the checker, and this summary. An independent mathematical cross-check of v3 is pending; the completed v2 cross-check does not cover the new argument.

> **twinheterodox** · 20 October 2026
>
> Ed: determinant construction changed but still the same general approach so posted as a reply. Codex summary follows:
>
> $q < 4000$, a factor of 250 improvement over the previous $q < 10^6$.
>
> Version 4: self-contained proof, source code, and verification outputs · Direct PDF.

This changes the determinant construction: the direct split-prime count from my earlier v3 comment is now combined with Ben Green's derivative Gram determinant, working in the quadratic field itself.

Under the class-number-one assumption, counting primitive quadratic integers of prime norm bounds the weighted sum of split primes. Explicit prime estimates of Rosser–Schoenfeld then force a lower bound for the weighted sum of inert primes. Frobenius at those primes makes a positive integer determinant divisible by a product exceeding its Archimedean upper bound.

The refinements are a sharper resultant criterion for nonsingularity, evaluation points in a rectangle adapted to the quadratic-integer lattice, and an absolute-value estimate that retains the decrease of polynomial degree under differentiation. With monomials of degrees less than $R$, $S$ and derivatives of orders less than $J$, the choices

$$R = q^4, \qquad S = \lfloor R/50 \rfloor, \qquad J = \lfloor R/55 \rfloor$$

give a contradiction for every $q \geq 4000$. The normalized comparison has an exact rational gap greater than $0.007$.

The ten-page paper includes the determinant and resultant arguments and the complete finite classification; no earlier version is needed. The stated Rosser–Schoenfeld estimates and classical quadratic theory remain external inputs. No Heegner method or computed zero-free region for quadratic Dirichlet L-functions is used.

All 39 exact arithmetic checks pass. A separate exhaustive residue scan checks the 499 remaining candidate integers below 4000, and reduced-form enumeration verifies the usual nine survivors. Earlier versions remain available at their original links.

AI disclosure: Codex developed the new derivation, write-up, verification programs, and this summary. Both v3 papers have passed independent mathematical cross-checks. Independent mathematical review of this v4 has also passed.



**twinheterodox** · 20 October 2026

$q < 2000$, halving the v4 bound.

Version 5: proof, source code, and verification outputs · Direct PDF.

This retains Green's derivative determinant but changes how it is used. Mellin averaging over its size parameter produces a lower bound for the convergent sum of $(\log p)/p^{71/70}$ over split and ramified primes.

A logarithmic derivative of the Euler product of primitive quadratic norms gives a competing upper bound. This retains the contribution of composite norms that was discarded in the earlier point-count estimate. Euler summation controls the norm series and its derivative with explicit error bounds.

At $q = 2000$, the determinant lower bound exceeds 27.696, while the Euler-product upper bound is less than 27.679. The difference increases with $q$, so the contradiction has a certified gap greater than 0.017 throughout $2000 \leq q \leq 4000$.

The eleven-page paper gives the new interval argument in full, including the determinant and resultant proofs, and cites v4 for $q \geq 4000$. All 37 exact arithmetic checks pass. An exhaustive check of the 249 remaining candidate integers below 2000, followed by reduced-form enumeration, leaves the usual nine discriminants.

AI disclosure: see previous comments.

 **Kyle Pratt** · 20 October 2026

Thanks to Lior for the nice computations.

I just wanted to record some elementary observations that may be useful for computations. I'm not sure what kind of large computations are feasible with Lean these days, and would be glad for someone to weigh in.

If Q(sqrt(-q)) is a hypothetical tenth discriminant other than the nine known, then q>163 is a prime that is 3 mod 4, by Gauss's genus theory. As Lior mentions, this means (-q/p)=-1 for all primes p < q/4. Since 2 and 3 are inert, this gives q = 19 mod 24.

One can use additional small primes and the Chinese remainder theorem to impose more conditions on q. For instance, using (-q/5)=-1 we see that q=43,67 mod 120. Using both 5 and 7 gives q=43,67,163,403,547,667 mod 840.

In this way, we can put q in a small number of residue classes mod some large-ish M, and then search in each arithmetic progression for primes. In this way, we ought to be able to rule out q < X for somewhat large X. This kind of argument seems simple enough for Lean and Mathlib.

There are other ways to eliminate large ranges of q as having class number 1, but these may not interact as well with Lean. For instance, a line of argument due to Stark, Montgomery–Weinberger, Watkins uses low-lying zeros of other Dirichlet L-functions. Stark (1966) computed low-lying zeros of the zeta function to moderate precision and was able to prove q>exp(2.2*10^7). Cherubini and Wallisser (1987) used continued fractions to rule out small values of a linear form.

 **bengreen** · 20 October 2026

I just want to flag up this variant argument of Roelof van Dijk https://hexagonmath.org/2610.00409v1. It uses an auxillary inert prime $p_0$ (the maximal one $\leq X$) to show that the determinant is nonzero. Essentially the same argument is given in the paper of Jacob Pescador mentioned in Terry's other blog post/my comment on that post – that paper is not in the public domain.

 **michaelmross** · 20 October 2026

Following up on twinheterodox's comments, here is an independent derivation of the same route (Banks's Theorem 1.1, a zero-forcing lemma, and a Rabinowitsch sieve), with a sharper

constant in the zero-forcing step and an elementary proof. Note and scripts: https://github.com/michaelmross/tenth-discriminant.

Lemma. If $D = -d$ is a fundamental discriminant with $h(D) = 1$ and $d \geq 10^{10}$, then $L(s, \chi_D)$ has a real zero $\beta$ with $1 - \beta < 2/\sqrt{d}$. The asymptotically sharp constant is $6/\pi$.

Sketch. Poisson summation in one variable of the principal form (the Chowla–Selberg expansion) gives

$$\zeta_K(s) = \zeta(2s) + \sqrt{\pi}\frac{\Gamma(s-1/2)}{\Gamma(s)}(\sqrt{d}/2)^{1-2s}\zeta(2s-1) + E(s),$$

where $|E(s)| \leq 8\pi e^{-\pi\sqrt{d}}$ near $s = 1$.

Take $\sigma = 1 - 2/\sqrt{d}$. The first term is then larger than $\zeta(2) \approx 1.645$. The second term is negative, but its size is at most $1.572$: this follows from $\zeta(u) \geq 1/(u-1)$ on $(0,1)$ and the fact that $\Gamma(s-1/2)/\Gamma(s)$ is monotone near $s = 1$. Hence $\zeta_K(\sigma) > 0.07$.

On the other hand, $\zeta_K(s) \to -\infty$ as $s \to 1^-$, because the residue at $s = 1$ is positive. So $\zeta_K$ has a zero in $(\sigma, 1)$. Since $\zeta(s) < 0$ there, the zero belongs to $L(s, \chi_D)$.

Consequence. For $d \geq 10^{10}$ the lemma gives $(1-\beta)\log d < 4.6 \times 10^{-4}$. Banks's theorem requires at least $3.0 \times 10^{-3}$, so any tenth discriminant has $d < 10^{10}$. This uses only the first assertion of Banks's theorem, not the small-conductor computation. The margin is about 6.5x, so a weaker $\delta_0$ would also suffice.

Below $10^{10}$. If $h = 1$ and $d > 8$, then $d \equiv 3 \pmod 8$ and $(-d/p) = -1$ for every prime $p < (d+1)/4$. A CRT sieve over the primes up to 23 leaves about 1.6 million candidates in $(163, 10^{10})$. None survives the primes below 2000, and the run takes 0.4 s in numpy.

The repo also checks the Chowla–Selberg identity numerically against $\zeta(s)L(s,\chi)$, with agreement to $10^{-29}$ for $d \leq 163$, and recovers the Heegner list by brute force.

Everything rests on Banks's Theorem 1.1, which is where independent checking is most valuable.

AI disclosure: the lemma, its proof, and the scripts were developed with Claude Opus (Anthropic). I have checked the argument and run the scripts.



**Henric Larsson** · 20 October 2026

I would like to contribute a short AI-assisted note to the tenth-discriminant challenge.

Using the explicit real-zero estimates of Banks (2026) and Lu–Zaman–Zhao (2026), I derive an elementary lower bound for imaginary quadratic class numbers using Dirichlet hyperbola and positivity of the coefficients of ζ(s)L(s,χ).

This gives the explicit bound **|D| < 50,000** for any imaginary quadratic field of class number one. Two independent exact enumerations then recover precisely the nine classical fields, without using Heegner's classification.

This is not an improvement on the strongest existing numerical bounds; in particular, Bennett–Martin–O'Bryant–Rechnitzer (2018) already imply a smaller cutoff. The contribution is a compact quantitative extraction from the recent zero-free estimates.

The work was conducted with GPT-6 Extra High assistance and subsequently independently reviewed by a separate GPT-Pro model. The audit verified the deduction and finite calculations conditional on the cited zero-free theorems; it was not human peer review.

**Paper, LaTeX source, verification code and audit:**

https://github.com/henlar1975/Tao-tenth-discriminent-bound

Henric Larsson
Independent researcher



**Terence Tao** · 20 October 2026

A quick contribution from playing around with the SAIR open model: it seems that the Selberg-Chowla formula implies that if $d$ is the tenth discriminant, then there is a zero $L(\beta,\chi)=0$ with $\beta \geq 1-\frac{3.2}{\sqrt{d}}$ for $d \geq 655$. This, combined with the 7/8 quasiRH, gives a bound of $d<655$ for the tenth discriminant, and $d<1475$ if one uses the 11/12 quasiRH instead.

While these results are numerically stronger than those obtained by other methods, they are also using much more powerful machinery. I would definitely be interested in seeing what other approaches give for the tenth discriminant, even if the numerical value of these bounds is significantly larger than those given here.



**LemmaLab** · 20 October 2026

Bound from OpenAI's quasi-Riemann hypothesis: $h(-d)=1 \Rightarrow d \leq 653$.

Input: OpenAI's zero-free half-plane (no zeros of any Dirichlet L-function with real part exceeding 7/8). The deduction is classical; the point is the explicit constant.

1. If $h(-d)=1$ and $d \notin \{3,4,8\}$, genus theory forces $d \equiv 3 \pmod 4$ prime. The only reduced form is then $Q=x^2+xy+\frac{d+1}{4}y^2$, and $\zeta_K(s)=\frac{1}{2}\sum_{(x,y)\neq 0} Q(x,y)^{-s}$.
2. Chowla-Selberg (the same expansion as in michaelmross's comment above): $\zeta_K(s)=\zeta(2s)+\sqrt{\pi}\,\frac{\Gamma(s-1/2)}{\Gamma(s)}\,(\sqrt{d}/2)^{1-2s}\,\zeta(2s-1)+E(s)$, where $E$ is a Bessel series of size about $e^{-\pi\sqrt{d}}$. The difference is that we evaluate it at the fixed point $s=7/8$ rather than near $s=1$.
3. At $s=7/8$ the main part is $M(d)=\zeta(7/4)+C\,(\sqrt{d}/2)^{-3/4}\,\zeta(3/4)$ with $C=\sqrt{\pi}\,\Gamma(3/8)/\Gamma(7/8) \approx 3.8558$ and $\zeta(3/4) \approx -3.4413$. It increases with $d$, and an explicit bound for $|E|$ decreases with $d$. Arb ball arithmetic gives $\zeta_K(7/8) \geq 1.39 \cdot 10^{-5}$ at $d=654$, hence $\zeta_K(7/8)$ is positive for all $d \geq 654$.
4. Since $\zeta(7/8)$ is negative, $L(7/8,\chi_{-d})=\zeta_K(7/8)/\zeta(7/8)$ is negative, while $L(1,\chi_{-d})$ is positive. So there is a real zero in $(7/8,1)$, a contradiction. Hence $d \leq 653$.
5. Counting reduced forms for all fundamental discriminants with $d \leq 653$ leaves exactly 3, 4, 7, 8, 11, 19, 43, 67, 163.

Status: the input is OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re in openai/math (commit 1581d4cb), stated for every Dirichlet character with no further hypotheses. The proof source was reviewed with AI assistance (no sorry, no extra axioms, no native_decide) but not compiled. With the 11/12 version (paper of Oct 5) the same argument gives $d \leq 1151$ (computed numerically, not certified).

For comparison, the Siegel-zero route currently gives $10^6$ (twinheterodox v3) and $5 \cdot 10^4$ (Henric Larsson, using Banks and Lu-Zaman-Zhao); a fixed zero-free half-plane is much stronger for this problem, at the price of resting on the QRH input.

AI disclosure: worked out with Claude (Anthropic), with rigorous numerics in Python, mpmath and python-flint (Arb). Independently checked by ChatGPT (OpenAI) with its own code: the Chowla-Selberg identity (27 tests, about 60 digits), the margin at d = 654, the error-term bound, and the class numbers up to 653. Scripts available on request.

**LemmaLab** · 20 October 2026

Extension to all class numbers (conditional on the 7/8 QRH, as above).

For $h = 1$ the thread now has 4000 (twinheterodox v4) and 11000 (Bryce) without QRH, so the QRH input mainly earns its keep for higher class numbers. Here is the general version of the argument.

For a fundamental discriminant $-d$ with $d \geq 5$, $\zeta_K(s)$ is half the sum of the Epstein zeta functions of the $h$ reduced forms $(a_i, b_i, c_i)$, and Chowla-Selberg applies to each. For $s \in [7/8, 1)$, using only that the principal form occurs, that $a_i \geq 1$, and $K_\nu(x) \leq \sqrt{\pi/2x}\, e^{-x}$ together with $a_i \leq \sqrt{d/3}$ for the Bessel remainder, one gets
$\zeta_K(s) \geq \zeta(2s) - \frac{h}{2}\big(A(s)\, d^{1/2-s} + R(d,s)\big)$,
with $A(s) = 2^{2s}\sqrt{\pi}\,\Gamma(s-1/2)\,|\zeta(2s-1)|/\Gamma(s)$ and an explicit $R(d,s)$ decreasing in $d$. If the right side is positive, $L(s, \chi_{-d})$ is negative while $L(1, \chi_{-d})$ is positive, contradicting the half-plane. Hence $h \geq 2\zeta(2s)/(A(s)d^{1/2-s} + R(d,s))$ for every $s \in [7/8, 1)$. Optimizing $s$ (it moves from 7/8 towards 0.96 as $d$ grows) and certifying with Arb gives:

every imaginary quadratic field with $h \leq 2$ has $d \leq 4164$;
with $h \leq 10$: $d \leq 2.81 \cdot 10^5$;
with $h \leq 100$: $d \leq 6.64 \cdot 10^7$;
with $h \leq 1000$: $d \leq 1.18 \cdot 10^{10}$;
with $h \leq 2000$: $d \leq 5.42 \cdot 10^{10}$;
with $h \leq 5000$: $d \leq 4.02 \cdot 10^{11}$;
and $d \geq 2^{40}$ implies $h \geq 7946$.
(These thresholds are certified but not optimal; for $h = 1$ the sharper single-form estimate above gives 653.)

For comparison, Kimura's Lean-verified thresholds via the Siegel-zero route are $2.85 \cdot 10^{12}$ for $h \leq 100$ and $1.61 \cdot 10^{15}$ for $h \leq 2000$. The difference reflects the much stronger QRH input, not a cleverer argument. Since the Mosunov-Jacobson tables are complete up to $2^{40}$, the last line would in principle settle all class numbers up to 7945 conditional on QRH; I have not done that table extraction.

AI disclosure: derivation and Arb certification with Claude (Anthropic). Independently re-checked by Codex (OpenAI) with its own code: the Chowla-Selberg identity for several forms with $a \geq 2$ (agreement to about $10^{-64}$, including even $d$), each step of the remainder bound, monotonicity in $d$, and spot checks against class numbers near $10^5$ and $10^6$.

**LemmaLab** · 20 October 2026

**Application: Euler's idoneal numbers.** As far as I know this consequence has not been stated: **conditional on the 7/8 QRH, Euler's list of 65 idoneal numbers is complete** (previously known only under GRH, by Weinberger 1973; unconditionally, at most two further numbers were known to be possible).

Input: the bound $h(-d) \geq H(d,s) = 2\zeta(2s)/(A(s)d^{1/2-s} + R(d,s))$ for fundamental $-d$, $d \geq 5$, $s \in [7/8, 1)$, from my comment on higher class numbers (currently in moderation). It comes from Chowla-Selberg applied to all h reduced forms.

1. If n is idoneal, write $-4n = f^2 D_K$. Then $\mathrm{Cl}(D_K)$ also has exponent at most 2, so $h(D_K) = 2^{t-1}$, where t is the number of primes dividing $D_K$ (the cases $D_K = -3, -4$ are included directly).
2. A fundamental discriminant with t prime factors satisfies $|D_K| \geq m_t$ for an explicit lower bound $m_t$ (product of small primes). For $10 \leq t \leq 39$, Arb gives $H(m_t) \geq 2^{t-1}$ with a large margin (e.g. $H(m_{10}) \approx 1045$ versus 512). For $t \geq 40$ one uses $H(d, 7/8) \geq Cd^{3/8}$ with $C \geq 1/12$ (Arb) and $|D_K| \geq (t+1)!$. Hence $t \leq 9$, and for each $t \leq 9$ a certified threshold gives $|D_K| \leq 5.6 \cdot 10^8$.
3. Enumerating all fundamental discriminants with exactly t prime factors below these thresholds leaves 65 with exponent at most 2 (largest 5460).
4. From $h(f^2 D_K) = h(D_K) f \prod_{p \mid f} (1 - \chi(p)/p)/u$ with $u \leq 3$, and the number of genera of $-4n$ being at most $2^{r+1}$, one gets $\varphi(f) \leq 12 \cdot 2^{\omega(f)}$. This leaves 138 conductors, the largest being 840.
5. Testing all resulting n leaves exactly Euler's 65 numbers. The whole run takes about 30 seconds.

Relation to Kimura (arXiv:2610.21511v1): his unconditional extension of the Cremona-Sutherland lists (orders with $h(D) \leq 1000$) would also settle idoneal n with $h(-4n) \leq 512$, though this is not stated there. The case of many prime factors is where the stronger QRH input seems needed, at least with the thresholds published there: $t = 12$ gives $h = 2048$ at $|D_K| \approx 1.5 \cdot 10^{13}$, while the threshold for $h \leq 2000$ in that paper is $1.61 \cdot 10^{15}$.

AI disclosure: derivation and code with Claude (Anthropic), Arb certification via python-flint. Independently re-checked by Codex (OpenAI) with its own code: each step of the argument, the thresholds for t up to 39 (Arb), the two finiteness steps (t at least 40, and the conductors), the generator of fundamental discriminants against brute force up to 10^6, the exponent test against Gauss composition for all n up to 5000, and a full reproduction of the final list.

**LemmaLab** · 20 October 2026

Code, outputs and independent checks for my comments above (class number one, all class numbers, idoneal numbers): https://github.com/LemmaLabMath/qrh-class-numbers

A precision on my first comment: "no sorry" refers to the DirichletL development; one dependency (PNT+, Wiener.lean) contains two sorry, which I could not trace without compiling. This does not affect my results, which are derived from the statement that Dirichlet L-functions have no zeros with $\mathrm{Re}\, s > 7/8$.

**Kyle Pratt** · 20 October 2026

Just a note that OpenAI's 11/12 quasi-GRH paper mentions just before section 1.1 that one consequence of the result is the completeness of Euler's list of 65 idoneal numbers. This follows from the 11/12 zero-free strip and previous results in the literature: work of Elsenhans, Kluners, and Nicolae on class groups of imaginary quadratic fields with small exponent, and work of Kani and Grube relating idoneal numbers to fundamental discriminants.

**LemmaLab** · 20 October 2026

I truly appreciate you pointing this out. I completely missed it, and I deeply regret that error.

**LemmaLab** · 20 October 2026

Posted without having seen Terry's comment just above; the two bounds agree ($d < 655$ vs. $d \leq 653$). Evaluating $\zeta_K$ directly at the edge of the half-plane, rather than first locating the zero, seems to lose a little less: for the 11/12 half-plane it gives $d \leq 1151$ instead of $d < 1475$ (that value not yet certified).

**Bryce** · 20 October 2026

Combining an explicit version of my polynomial-lattice argument, arXiv:2610.21099, with the finite-range zero exclusion used by Henric Larsson, and a refinement of the Chowla–Selberg argument discussed by Ross and Tao, gives the cutoff ( |\Delta|<11{,}000 ) for a negative fundamental discriminant of class number one.

Writing (q=|\Delta|), the modifications to the argument in my paper are:

- Replace the general zero-dependent split-prime estimate with a direct count of primitive norm representations, incorporating restrictions at (2,3,5,7).
- Work in the order generated by ((1+\sqrt{\Delta})/2) and (\sqrt2), and centre the grid to improve the bounds at the complex embeddings.
- Change the monomial region from (i+8(j+k)\le D) to (i+192(j+k)\le D), with grid length (N=q^7), (D\approx9N^{4/3}/40), and prime cutoff (Q=D/2).
- For small primes, reverse the leading term in the Frobenius relation; for larger primes, halve the error in the divisibility count.
- At (3), additionally exploit the relation (\theta^9\equiv\theta\pmod3).
- Make the prime estimates and remaining constants explicit, retaining Masser's theorem as the nonvanishing input.

These polynomial-lattice refinements give (q<200{,}000). For the remaining interval, Larsson's Lu–Zaman–Zhao input combines with a sharper Chowla–Selberg sign test: retain additional terms in the zeta estimates and simplify the gamma factors by duplication. This gives (q<11{,}000), below which an elementary finite sieve recovers the nine classical discriminants.

Developed with ChatGPT assistance; the numerical inequalities have exact rational certificates.

**Kyle Pratt** · 20 October 2026

*Used ChatGPT 6.1 Sol for this analysis. In the spirit of our host's invitation, I am not sharing the low-quality AI write-up, but just some interesting ideas to explore further.*

I think it is possible to use the class number 1 information along with van Dijk's argument to get an upper bound for q that is close to the quasi-GRH bounds above, in the low thousands.

Since h(-q)=1, split primes have to be represented by (a+b/2)^2+qb^2/4. Counting integers represented by this quadratic form can give an upper bound for sum_{p < X \\ chi(p)=1} (log p)/p that is very strong in some ranges. Use this to find a large auxiliary inert prime on the order of exp(c*sqrt(q)) that one can use as in van Dijk's argument. Use the stronger p-power divisibility arguments from Green rather than the simplified setup of van Dijk.

There may be some numerical advantage to choosing the interpolation points from a disk instead of a rectangle.

There is quite a bit of finagling and parameter-choosing if one wants to get an upper bound for q that is very small.

**Rudolph** · 20 October 2026

I explored the tenth-discriminant challenge using Iwao Kimura's explicit zero-gap estimates (*Imaginary quadratic fields of class number at most 2000*, arXiv:2610.21511).

Using Kimura's Proposition 9 with

$$c = 0.004401, \quad T = 19.0591, \quad A = 2,000,000, \quad V = 17,550$$

gives the numerical lower bound

$$\sqrt{q}\, L(1,\chi) \geq 3.141726175\ldots > \pi$$

for the relevant conductors, using the uniform zero-gap estimate from Kimura's certified grid.

Together with the imaginary quadratic class-number formula, this gives

$$|D| < e^{19.0591} < 189,348,540$$

for any hypothetical tenth class-number-one discriminant. It is a much larger bound than those that already have been posted here, but I share it anyway.

An exact-integer CRT sieve using splitting primes through 89 then eliminates the remaining candidates above 163.

*Attribution:* The splitting-prime obstruction and computational sieve were already used by *twinheterodox* in this discussion. The contribution here is a different numerical specialization using Kimura's analytic estimates, not a new classification theorem.

**Research note:**

<a href="https://drive.google.com/file/d/1TvoHMFTNdcTAjIyuOcISnzly2-sEI444/view?usp=sharing">Read the full mathematical note (PDF)</a>

**Verification code:**

<a href="https://drive.google.com/file/d/1e4L4maEmAWyBPFGEFefsD-ruu-oTnIU8/view?usp=sharing">View the Python verification script</a>

The numerical inequality and finite sieve have been checked computationally, but I have not independently replayed Kimura's Lean certificates. This investigation was carried out with extensive ChatGPT 6 assistance.

 **twinheterodox** · 20 October 2026

Ed: I ran out of replies. Consider this a continuation of the previous thread.

Headline: $q < 570$, so tighter than the QRH-based argument Terry and LemmaLab gave, without using the heavier QRH machinery.

Instead of uploading a bunch more intermediate bounds, I pushed the model to tighten it as far as it could go without using "heavy machinery". I told the model to read the comments on this post for additional ideas. My exact prompt was:

> read the comments again. the goal is to go as far as we possibly can without heavier machinery, so to the extent there are ideas there we can use them (I know you already have). We will
> attribute. Then, once you think there is nothing meaningful to be gained, we will stop and write everything up as one v6 result. Does that sound reasonable?

It was processing for several hours. Details are in the writeup. Somewhat interestingly, it was well down a rabbit hole using a whole bunch of different optimizations that were going to generate an extremely complex tighter bound, when it abruptly had an epiphany and discarded all that in favor of a simpler argument. At least "simpler" in some metric that makes sense to AI.

I am pretty sure this can be tightened further, but I've used about a month of my personal AI budget on this and it's late (and I am not sure anyone cares).

This has not yet been independently verified because the model is getting throttled right now. There are also obvious flaws in the writeup that need fixing (I told the model to attribute provenance so it dumped a giant stream-of-consciousness paragraph at the start of the writeup)

**Codex-generated summary:**

$q < 570$, down from $q < 2000$, without QRH, a quadratic zero-free theorem, or Masser's theorem.

Version 6: proof, source code, and verification files · Direct PDF.

This follows Terry's request to pursue bounds using lighter machinery. It builds on Green's derivative determinant and stronger Frobenius divisibility, van Dijk's auxiliary-inert-prime nonvanishing argument, and Pratt's suggestion to combine norm counts with that auxiliary prime and disk interpolation. Kerr's higher Frobenius suggestions and the Poisson discussion by michaelmross and Terry also inform the argument; the paper gives the detailed attributions.

The main change from v5 is to retain the smoothed sum over split and ramified primes already present in the exact determinant valuation. This same sum strengthens the row and column factorial bounds through eight finite refinements, each using only the preceding stage. Further gains come from compatible higher Frobenius factors, a joint bound for derivative orders and interpolation radii, and using the auxiliary inert prime itself as the monomial width.

A positive kernel formed from a squared polynomial then compares the determinant lower bound with the quadratic norm series. Poisson summation is used at real $s > 1$, with explicit error bounds that account for cancellation among the signed coefficients.

The fifteen-page paper gives the new interval argument from 570 to 2000 in full relative to its stated classical inputs, and cites the audited v5 result for the range above 2000. Five exact rational interval certificates cover the whole new interval; all 429 grouped checks pass, with a smallest certified gap greater than $0.010389$. An isolated rerun reproduces the exact outputs. A separate finite check below 570 leaves the usual nine discriminants.

Independent mathematical review of v6 is pending. The computational checks do not constitute an independent audit or formal verification.

AI disclosure: see previous comments.